Literature Survey and Research Gap

Project: Self-Healing Beyond Liveness: Model-Aware Recovery, Delivery and Failure Prediction for ML Forecasting Services on Kubernetes. Summaries are based on each paper's abstract and first pages. Years marked ~ are inferred from the PDF and must be verified against the IEEE record before citing.

#	Paper	Authors	Year	What it states	Research gap	How our project differs
1	AI-Augmented Chaos Testing: Toward Intelligent Resilience Engineering in Cloud-Native Systems	S. Nigam, S. Patel	2026	GNN criticality analysis + transformer anomaly forecasting + RL to orchestrate chaos experiments, with a digital-twin 'Resilience Mirror'; up to 46% fewer SLO violations and 38% lower MTTR on Kubernetes microservices.	Generic microservice/IoT/GPU workloads; injects faults but does not treat ML model state (loaded/correct) as part of health or recovery.	We inject ML-specific faults (corrupt model = silent NaN output, OOM during model load, hang) on a forecasting service and add model-aware recovery.
2	A Comparative Study of Automatic Migration from Monolith App to Microservices	J. Zairi, A. Kenzi	2025	Systematic review of 20 studies on automatic monolith decomposition; proposes a framework and evaluation criteria.	Review only: no deployed system, no runtime performance or fault-recovery data.	We migrate a real SARIMA+LSTM app and measure latency, recovery and failures for monolith vs Compose vs Kubernetes.
3	A Predictive Elastic Scaling Method for Kubernetes Based on Load Forecasting	L. Li	2026	PathFormer forecasts CPU utilisation; replica counts are pre-computed and applied through Prometheus + the K8s API; outperforms native HPA.	Autoscaling for generic load; assumes a new pod is instantly useful (ignores model-load/warm-up time).	Our scaling experiment (secondary) accounts for model warm-up lead time and compares against HPA baselines.
4	A Stacking Ensemble ML-Based Failure Prediction Model for Optical Networks with Imbalanced Data	Z. Sun et al.	2023	Stacking ensemble predicts failures when only 5.3% of samples are failures; F1 = 98.97%.	Optical-network telemetry; not containers or ML services.	Supports our handling of rare failures (class weights/SMOTE, report F1); we predict container/model failures from pod + forecast-quality signals.
5	AdaptiPID: Model-Aware Adaptive Gain Tuning with Anti-Windup Compensation for Robotic Manipulator Control	Y. Zhou	2026	Adaptive PID with anti-windup and joint-coupling compensation for robot arms.	Unrelated domain (robot control).	Not relevant; excluded from related work.
6	Architecture-based Failure Prediction via LSTM and Bayesian Network in Service-oriented Systems	X. Wu et al.	2022	LSTM predicts component reliability time series; a Bayesian network infers impending failures; +14% accuracy.	Component reliability series only; no output-quality or model-state signals; no recovery action.	Our predictor fuses pod telemetry with forecast-quality signals and triggers proactive replacement through the operator.
7	Autoencoder-based Anomaly Detection in Microservices using Distributed Tracing (AnoTraceAE)	S. Shahini, H. Momeni	2024	Convolutional autoencoder over span/trace embeddings; evaluated on TrainTicket.	Detection only; benchmark demo app; no ML-model health, no recovery.	We detect model-level failures (silent bad forecasts) and close the loop with recovery.

8	Chaos Engineering based Kubernetes Pod Rescheduling through Deep Sets and Reinforcement Learning	M. Zaccarini, F. Poltronieri, D. Borsatti, W. Cerroni, L. Foschini et al.	2025	TELKA scheduler: chaos engineering + RL + digital twin to reallocate pods evicted by faults; Deep Sets removes the fixed-node-count limit.	Pod placement after node faults; not application-level correctness or model loading.	We study application-level silent failures and model-load recovery, not scheduling.
9	Deployment of Seasonal Autoregressive Integrated Moving Average (SARIMA) Models for Network Reliability Prediction	D. Singh	2024	Step-by-step deployment of SARIMA to forecast short-term network reliability.	Model use only; no containerisation, probes, recovery or operational metrics.	We deploy SARIMA and LSTM as services and study their operational reliability (MTTR, availability).
10	Evaluating operational readiness using chaos engineering simulations on Kubernetes architecture in Big Data	G. Siwach, A. Haridas	2022	Random pod termination in a Big Data K8s setup; infers pod recovery time.	Only pod kills; no ML model state, no silent faults, no statistical experiment design.	Four-fault taxonomy, repeated trials, and a DoE/RSM MTTR model with model-load decomposition.
11	Is it Worth Migrating a Monolith to Microservices? An Experience Report on Performance, Availability and Energy Usage	V. Berry et al.	2024	Migrates a web app; MSA wins without replication, monolith wins when the whole app is replicated; also studies energy.	Generic web app; no ML models or fault-recovery behaviour.	Same A/B/C comparison on an ML app, extended with fault injection and model-aware recovery.
12	Kube-CC: Building an Experiment Education Platform for Big Data Courses Based on Kubernetes	F. Xue et al.	2023	K8s + Go platform for one-click Big Data cluster deployment and recovery in teaching labs.	Education platform; no resilience measurements.	Background on Kubernetes use only.
13	MicroMILTS: Fault Location for Microservices Based on Mutual Information and LSTM Autoencoder	L. Yang et al.	2022	BIRCH anomaly detection, then LSTM-autoencoder reconstruction deviations + mutual information to locate the faulty service.	Localises faults from infrastructure metrics; no model-quality signals; no recovery.	We add forecast-quality signals and automatic recovery.
14	Multi-Level Ensemble Anomaly Detection for Microservices: Statistical Analysis, Isolation Forest and Adaptive Thresholding	R. Gangwar, V. Srivastava, P. Srivastava	2026	Prometheus metrics through statistical profiling + Isolation Forest + adaptive thresholds, with a dashboard; ERP-like testbed.	Metric-only detection on a synthetic ERP; no ML inference service, no healing.	Isolation Forest is used on pod telemetry fused with output-quality signals and wired into recovery.
15	Recall-Prioritized Anomaly Detection in Cloud Microservices using Unsupervised Transformer Autoencoders	A. C. et al.	2026	LogBERT-Lite transformer autoencoder on OpenStack logs; high recall (8 of 11 injected faults) at 4.37% precision.	Log-based, low precision; no remediation.	We target silent ML faults and tie detection to recovery with a measured lead time.
16	Research on Constraint-Aware Fault Recovery Diagnosis Decision Method for Offline Edge Kubernetes Scenarios	Z. Wei, K. Bai	2026	CAR: Repair / Degrade / Rebuild decision tree; MTTR 127 -> 34 min in an offline ARM64 case study.	Manual-style diagnosis for constrained edge clusters; no ML service.	Our degrade tier (LSTM -> SARIMA -> climatology) is an automated, ML-service-specific analogue.
17	Residual Artifact Governance With ML-Enabled Garbage Collection Prediction in Kubernetes-Based 5G/6G Service-Based Architecture	L.-M. Tufeanu et al.	2026	Master-agent garbage collection with an ML predictor of time-to-collapse from residual artifacts.	Telecom core network; residual-artifact leaks, not ML service correctness.	Same idea of ML early warning, applied to forecasting-service failures.

18	Robust Hybrid Autoencoder-LSTM Framework for Real-Time Anomaly Detection in Multi-Cloud Environments	S. Oviya et al.	2025	Autoencoder + LSTM combined score; 94.2% accuracy and 5.6% false positives on CICIDS2017.	Network-intrusion datasets; no operational recovery.	LSTM autoencoder is used as a comparison detector on our own fault traces.
19	Service Level Objective (SLO) Observability with Splunk and Dynatrace in Microservices	V. P. K. Gummadi, L. S. Chilamkurti	2026	SLO/SLI-centred observability, error budgets, reduced alert fatigue.	Practice/commercial-tool paper; no experiments on ML services.	We implement SLOs, burn-rate alerts and error budgets for a forecasting service with Prometheus/Grafana.
20	The Multi-Modal Anomaly Detection Method for Microservices Based on VGAE-Transformer (VTMAD)	C. Shao, X. Zhou et al.	2025	VGAE + Transformer on multimodal data; reconstruction error on the adjacency matrix.	Heavy detection model on public datasets; no healing.	Lightweight, deployable detector plugged into an operator.
21	TraceDAE: Trace-Based Anomaly Detection in Microservice Systems via Dual Autoencoder	J. Li, S. Ying, T. Li et al.	2025	Service Trace Graph; GAT structure autoencoder + LSTM attribute autoencoder; detects response and invocation anomalies.	Trace anomalies only, offline datasets; no model-health or recovery.	Detection feeds real recovery and is evaluated on MTTR and failed requests.
22	Enhanced Multi-model Online Conformal Prediction	E. Hajihashemi, Y. Shen	2025	Conformal prediction with a candidate set of models: one model is selected per time step and a bipartite graph keeps only effective models, giving smaller prediction sets at lower compute.	Online-learning theory; measures set size and compute only; not used for health checks, fault detection or recovery in a running service.	Conformal coverage is used as a service-correctness gate with heterogeneous models (SARIMA, LSTM, climatology), tier fall-through and operator recovery. Possible upgrade: an online/adaptive envelope and model-subset selection for the fallback tier.
23	SGCAD: Structure-aware Granger Causality-based Anomaly Detection	Z. Zhang, C. Wan, B. Cao, L. Zhao	2026	Neural Granger causality learns temporal dynamics and metric dependencies on normal data; global prediction error, per-service anomalies and structural change give an anomaly score and a root-cause service ranking (RS-Anomic).	Detection and root-cause ranking from monitoring metrics only; no output-quality signal and no recovery action.	We use forecast-correctness signals and close the loop with recovery (restart, rollback, tier pinning).

Common gap across the surveyed papers

- They use generic workloads (TrainTicket, ERP-like apps, Online Boutique, network data); none treats ML model state as part of health.
- Silent failures (HTTP 200 with wrong forecasts) are not studied; probes and detectors only see infrastructure metrics.
- Detection and prediction are evaluated offline; few close the loop to automatic recovery with MTTR and failed-request metrics.
- No statistically designed (DoE/RSM) MTTR model that includes model-load time.

Novelties of this project

1. Semantic readiness/startup probe (canary inference with schema, NaN, climatology-bound and residual checks).

2. DoE/RSM MTTR model with decomposition into detect, schedule, image pull, model load and warm-up.

3. Forecast-quality-aware failure predictor (Isolation Forest / LSTM autoencoder on pod telemetry + output quality).

4. Model-artifact quality-gated progressive delivery (Argo Rollouts + Prometheus AnalysisTemplate).

5. ForecastService operator with tiered graceful degradation (LSTM -> SARIMA -> climatology -> cache).

Extras: SLO/burn-rate alerts, supply-chain signing (Trivy/cosign/Kyverno), open fault benchmark, SARIMA vs LSTM under identical faults.

Extensions added 2026-10-22 (measured; see TESTING_AND_KNOWN_ISSUES 3.25-3.27): random bit-flip fuzzing of the LSTM weights (12,600 trials; exponent-MSB flips make 50.2 % of answers wrong, the envelope misses 34 % of wrong answers, the golden check misses none); cosign-signed golden outputs with fail-closed readiness and an admission policy (80 cluster trials); operator rollback and escalation (bad release: restart-only 0/5, release-aware rollback 46 s, escalate 77 s). Prior art: exponent bit-flip damage is known (Bit-Flip Attack, Rakin et al., ICCV 2019); model signing with hash manifests exists (OpenSSF/Sigstore); metric-based automated rollback exists (Argo Rollouts); the claims here are the measurements and the semantic-health-driven escalation, to our knowledge. Papers 22 (2025, Asilomar) and 23 (2026, AANN) are summarised from their abstracts; years and author lists come from a Crossref lookup and should still be confirmed on IEEE Xplore.